Scan: **Claim:** If the sequence (z_n) converges to a limit b, where $b>0$, then it contains a **finite** no. of terms which are 0.

Proof: $\forall \varepsilon > 0, \varepsilon \in R, \exists N_1 \in N$ s.t. $\forall n \in N, n > N_1, |z_n - b| < \varepsilon \Rightarrow \boxed{b-\varepsilon < z_n < b+\varepsilon}$ (1)

- The equation (1) holds for all z_n, where $n > N_1$.
- Let $z_n = 0$ $\quad b-\varepsilon < 0 \Rightarrow b < \varepsilon$
  $0 < b+\varepsilon \Rightarrow -\varepsilon < b$ $\quad \boxed{-\varepsilon < b < \varepsilon}$ (Wrong)
  $\hookrightarrow \varepsilon > -b$

(Proof attempt failed)

Reader: Well, I select $\varepsilon = \frac{b}{2}$. There must be an integer N such that $|z_n - b| < \frac{b}{2}$ for all $n > N$. Obviously, all z_n (the whole "infinite" tail of the seq) will be positive. Consequently, the zero-terms of the sequence (z_n) may only be encountered among a **finite** number of the first N terms.
$\hookrightarrow \left(b - \frac{b}{2} < z_n < b + \frac{b}{2} \Rightarrow \frac{b}{2} < z_n < \frac{3b}{2}\right)$ (Very simple, but beautiful)

Note: This proof can be generalized for any arbitrary $\varepsilon > 0$.

Author: Excellent. Thus, the number of zeros among the terms of (z_n) can only be finite. If such is the case, one can surely drop these terms. Indeed, an elimination of any finite number of terms of a sequence does not affect its properties.
$\hookrightarrow$ Needs proof

For example, a convergent sequence still remains convergent, with its limit unaltered. An elimination of a finite number of terms may only change N (for a given ε), which is certainly unimportant.

Reader: It is quite evident to me that by **eliminating** a finite number of terms one does not affect the convergence of a sequence. But could an **addition** of a finite number of terms affect the convergence of a sequence?

Author: A finite number of new terms does not affect the convergence of a sequence either. No matter how many new terms are added and what their new serial numbers are, one can always find the greatest number N after which the whole infinite "tail" of the sequence is unchanged. No matter how large the number of new terms may be and where you insert them, the finite set of new terms cannot change the infinite "tail" of the sequence.